# Kavya Acharya

## FPGA Design Engineer

RESUME_1628

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Total Experience: 3 years

### PROFESSIONAL SUMMARY

FPGA Design Engineer with 3 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

### TECHNICAL SKILLS

Skills: UART, DSP, VHDL, Python, ILA, Verilog, Hardware Simulation, Xilinx Vivado, Tcl, Intel Quartus, AXI4, FPGA

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

### PROFESSIONAL EXPERIENCE

**FPGA Design Engineer | HelioNex Labs | Jul 2023 - Present**

- Verified 441 protocol corner cases with repeatable simulation suites
- Improved maximum operating frequency by 16% after timing-path optimization
- Lowered end-to-end data-path latency by 36% using pipelined logic
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

### PROJECTS

**High-Speed Data Logger | Verilog, Python, AXI4, Hardware Simulation**

- Created DMA and buffering logic with ILA debugging hooks.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

**Real-Time Signal Filter | VHDL, ILA, Verilog, FPGA**

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

### EDUCATION

B.Tech in Electronics and Communication Engineering, Silverlake School of Engineering, 2023
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

### CERTIFICATIONS

Digital Signal Processing Basics (2024)